Edge-AI Driven Elastic Mesh PSO for Low-Altitude UAV IoT Networks in Zero-Infrastructure Environments

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Abstract. UAV swarms searching a disaster area without infrastructure must relay victim reports to responders over their own links. SENTINEL treats the network as an elastic mesh that stretches and breaks during search and is re-formed only to deliver a confirmed report: Elastic-Mesh PSO (EM-PSO) plans waypoints, confirmed victims trigger delivery, weak detections get a second view, and an outage-based Signal-Spring keeps delivery links reliable. In about 17 000 simulated missions with a 2.4 GHz outage channel, 3–50 UAVs, six failure settings and nine tuned baselines, SENTINEL reports 23–29 % more victims within 60 s than the best baseline in urban rubble (2–7 % with a detector model fitted to rendered-frame detector outputs), and an always-connected mesh reports far fewer in time.

Keywords. UAV swarm, edge AI, particle swarm optimization, search and rescue, intermittent connectivity, low-altitude IoT

1. Introduction

After an earthquake, thermal-camera UAVs can detect people in rubble on board [1, 2], but a detection matters only once responders receive it over short-range links that buildings block, fading weakens and crashes cut. A connected swarm delivers at once but searches little, a spread-out one finds more but delivers by chance; we measure this trade-off as the share of victims reported within 60 s.

Our first version held each UAV within 50 m of its nearest neighbour, but a fixed radius ignores rubble, a close neighbour does not imply a connected network (Lemma 1), and its 25 m test area was smaller than the radio range. This revision contributes (i) *information-triggered delivery*: a UAV stops searching when a victim it confirmed has not reached the responder within a hold time T_h; (ii) *corroborative sensing*: weak detections get a lateral second view, fused so that a blocked angle cannot veto a clear one; and (iii) *EM-PSO with an outage-based Signal-Spring and link preservation*: EM-PSO plans the waypoints throughout, while its spring and link preservation act only during deliv-

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ery, as a safety property that keeps preserved links reliable (Proposition 1) and cuts link violations 37-fold; the reporting gains come from (i) and (ii). Code, data and extended tables: `https://github.com/KS-KARTHIK-05/SENTINEL` (branch v2).

2. Related Work

Connectivity controllers keep the graph connected through its algebraic connectivity λ_2 [3,4] or edge potentials [5], assuming a disk radio model. Intermittent connectivity schedules meetings [6], and age-of-information collection serves known sources [7], whereas victims are found by the search itself. Distributed PSO and Bayesian search [8,9] assume communication when needed, and MARL such as MAPPO [10] needs a trained policy. None combines online-discovered sources, a deadline and an obstruction-aware channel.

3. System Model

N UAVs fly at altitude h over tall no-fly, link-blocking buildings and low rubble that hides victims; a responder radio at the launch point is the sink. Each slot Δt every UAV broadcasts its position, victim ledger and acknowledgements, received at j with SINR

$$\gamma_{ij} = \frac{P_t G^2\, 10^{-(\mathrm{PL}(d_{ij})+n_{ij}L_o+X_{ij})/10}\,|h_{ij}|^2}{\sigma^2 + \sum_{k\in\mathscr{I}_j} \bar{P}_{kj}}, \tag{1}$$

with log-distance path loss $\mathrm{PL}(d) = \mathrm{PL}(d_0) + 10\eta \log_{10}(d/d_0)$, n_{ij} obstacles at L_o each, correlated shadowing X_{ij}, Rician fading (Rayleigh when obstructed) and colliding transmitters $\mathscr{I}_j$. Decoding needs $\gamma_{ij} \geq \gamma_{\mathrm{th}}$, so

$$P_{\mathrm{out}}(d,n) = \mathbb{E}_X\left[F_{|h|^2}\left(\gamma_{\mathrm{th}}/(\bar{\gamma}(d,n)\,10^{X/10})\right)\right], \tag{2}$$

and the reliable range is $R_\varepsilon(n) = \max\{d : P_{\mathrm{out}}(d,n) \leq \varepsilon\}$. The link budget (2.4 GHz, $P_t = 20$ dBm, $G = 2$ dBi, $B = 20$ MHz, noise figure 7 dB, $\gamma_{\mathrm{th}} = 5$ dB) leaves 113 dB of path loss before fading margins, and profiles follow published ranges [11,12]. In urban rubble a link is reliable to 91 m in the open and 48 m behind one obstruction (Table 1), so the old 50 m radius describes one obstructed link. With τ_k victim k's first confirmation and T_k its delivery time (∞ if never), the team maximises from local information

$$J = \frac{1}{|\mathscr{V}|}\sum_{k\in\mathscr{V}} e^{-T_k/\theta}, \qquad \theta = 300\,\mathrm{s}. \tag{3}$$

4. SENTINEL

Mode Switching. With $\mathscr{E}_i(t)$ the confirmed, undelivered entries UAV i owns (its own, and any older than $3T_h$, so a crashed UAV's copies still arrive), it enters DELIVER when $\max_{e\in\mathscr{E}_i(t)}(t-\tau_e) \geq T_h$ and returns to EXPLORE when $\mathscr{E}_i(t)$ is empty.

Table 1. Channel profiles and reliable ranges at 5 % outage with no and one obstruction (K: line-of-sight Rician factor; L_o: loss per obstruction).

Profile	η	σ_X [dB]	K [dB]	L_o [dB]	$R_{0.05}(0)$ [m]	$R_{0.05}(1)$ [m]
open	2	2	10	6	2434	684
suburban	2.5	4	6	10	339	95
urban rubble	3	6	3	6	91	48
severe	3.5	8	Rayleigh	8	35	21

EM-PSO. Each planning period a UAV picks the reachable waypoint that maximises

$$F_i(x) = w_c \hat{G}_i(x) + w_d \sum_{e \in \mathscr{T}_i} c_e e^{-\frac{\|x-x_e\|^2}{2s^2}} + \frac{w_g(\|x_i - g_i\| - \|x - g_i\|) - w_t \|x - x_i\|}{vT_p} - S_i(x), \tag{4}$$

with $\hat{G}_i$ expected new detections, $\mathscr{T}_i$ the tentative entries, g_i a frontier goal and w_t a travel cost. The Signal-Spring acts only while tethered (delivering or relaying):

$$S_i(x) = \lambda_i \sum_{j \in \mathscr{S}_i} \Big(\frac{[P_{\text{out}}(\|x - x_j\|, n(x, x_j)) - \varepsilon_i]_+}{\varepsilon_i} \Big)^2, \tag{5}$$

where $\lambda_i = \lambda_0(1 + \beta U_i)$, $\varepsilon_i = \varepsilon/(1 + \beta U_i)$ and U_i counts entries only i holds; it penalises outage, not distance. A constriction PSO ($\chi = 0.7298$, $c_1 = c_2 = 2.05$) [13] maximises F_i.

Link Preservation. $\mathscr{S}_i$ holds i's neighbours in the local minimum spanning tree of reliable links [14].

Lemma 1. *Nearest-neighbour distance constraints do not imply connectivity.*

Proof. Two UAV pairs 10 m wide and 190 m apart satisfy $R = 50$ m, yet no link joins them. □

Tethered UAV i scales its step u_i by the largest $\alpha \in [0, 1]$ such that $x_i' = x_i + \alpha u_i$ meets

$$\|x_i' - m_{ij}\| \le \tfrac{1}{2} r_{ij} - \tfrac{1}{2} v a_{ij}, \quad r_{ij} = R_\varepsilon\big(n^{\delta_{ij}}(x_i', \hat{x}_j)\big) \quad \forall j \in \mathscr{S}_i, \tag{6}$$

with $\hat{x}_j$ the last reported position of j (age a_{ij}), $m_{ij} = (x_i + \hat{x}_j)/2$, $\delta_{ij} = v(\Delta t + a_{ij})$ and n^δ counting obstacles inflated by δ.

Proposition 1. *If $j \in \mathscr{S}_i$, $i \in \mathscr{S}_j$, both UAVs are tethered, speeds are bounded by v, positions are at most a old and both steps satisfy Eq.* (6)*, link (i, j) is ε-reliable at $t + \Delta t$.*

Proof sketch. Midpoint estimates err by $\le va/2$, so Eq. (6) and the triangle inequality give $\|x_i' - x_j'\| \le \frac{1}{2}(r_{ij} + r_{ji})$; R_ε is non-increasing in the obstacle count, which the inflated count bounds. □

Table 2. Share reported within 60 s under six failure settings (urban rubble, 400 m, $N = 8$): none; one crash; 30 % of the swarm at once (simult.); three cascading crashes (casc.); Poisson crashes at twice the default rate; transmit-power brownout (brown.). Mean ± 95 % CI half-width over 30 seeds; best in bold; *: differs from SENTINEL ($p < 0.05$, paired Wilcoxon, Holm).

Method	none	single	simult.	casc.	Poisson	brown.
SENTINEL	**0.69±0.04**	**0.69±0.04**	**0.67±0.04**	**0.64±0.04**	**0.46±0.05**	**0.69±0.04**
B0 PSO	0.54±0.04*	0.52±0.04*	0.51±0.04*	0.51±0.04*	0.34±0.06*	0.54±0.04*
B1 first version	0.50±0.05*	0.50±0.05*	0.50±0.05*	0.49±0.05*	0.36±0.05*	0.51±0.05*
B2 comm.-aware PSO	0.56±0.03*	0.55±0.03*	0.52±0.03*	0.52±0.04*	0.34±0.03*	0.55±0.04*
B3 λ_2 control	0.53±0.03*	0.52±0.04*	0.51±0.04*	0.49±0.03*	0.34±0.05*	0.53±0.03*
B4 edge potentials	0.19±0.04*	0.16±0.03*	0.14±0.04*	0.16±0.04*	0.09±0.03*	0.18±0.04*
B5 MAPPO	0.50±0.04*	0.47±0.04*	0.45±0.04*	0.46±0.04*	0.33±0.05*	0.49±0.04*
B6 sweep	0.52±0.04*	0.51±0.04*	0.49±0.04*	0.49±0.04*	0.35±0.05*	0.52±0.04*
B7 hard wall	0.28±0.05*	0.28±0.05*	0.28±0.05*	0.27±0.04*	0.23±0.04*	0.29±0.05*
B8 relay chain	0.53±0.05*	0.52±0.05*	0.51±0.05*	0.48±0.05*	0.36±0.05*	0.53±0.05*

Table 3. Share reported within 60 s vs. swarm size (400 m, urban, Poisson failures); statistics as in Table 2.

Method	$N = 3$	$N = 5$	$N = 10$	$N = 20$	$N = 50$
SENTINEL	**0.20±0.04**	**0.39±0.06**	**0.67±0.04**	**0.75±0.03**	**0.79±0.03**
B0 PSO	0.15±0.03*	0.29±0.05*	0.53±0.04*	0.62±0.03*	0.66±0.04*
B1 first version	0.11±0.03*	0.23±0.05*	0.50±0.05*	0.65±0.03*	0.71±0.04*
B2 comm.-aware PSO	0.15±0.04*	0.31±0.05*	0.53±0.04*	0.64±0.04*	0.69±0.03*
B3 λ_2 control	0.14±0.04*	0.27±0.05*	0.53±0.04*	0.64±0.04*	0.67±0.03*
B4 edge potentials	0.02±0.01*	0.03±0.02*	0.24±0.05*	0.49±0.05*	0.71±0.04*
B5 MAPPO	0.15±0.04*	0.27±0.04*	0.50±0.05*	0.63±0.03*	0.73±0.03*
B6 sweep	0.15±0.04*	0.27±0.05*	0.44±0.04*	0.49±0.04*	0.62±0.03*
B7 hard wall	0.09±0.03*	0.15±0.04*	0.36±0.04*	0.62±0.05*	0.72±0.03*
B8 relay chain	0.10±0.03*	0.27±0.05*	0.52±0.04*	0.66±0.04*	0.75±0.02*

Corroborative Sensing. An entry with $0.40 \le c_e < 0.70$ requests the nearest neighbour's view ±15 m across the line of sight, which at $h = 30$ m looks 26.6° off nadir past a slab edge at 12 % more slant range. With $c_{e,v}$ the best confidence per sector v, SENTINEL fuses

$$\operatorname{logit} c_e = \max_v \operatorname{logit} c_{e,v} + \sum_{v \neq v^\star} [\operatorname{logit} c_{e,v}]_+, \tag{7}$$

with $v^\star$ the best view; weak blocked views cannot veto it, unlike naive-Bayes fusion.

5. Evaluation

Setup. One library runs in a fast simulator and in ROS 2 Humble nodes flying PX4 x500 quadrotors in Gazebo Harmonic through Micro XRCE-DDS, with YOLO26n running on board, both dropping packets by sampled SINR. Each run generates an area from its seed (25 victims, rubble, warm debris) and lasts 600 s; by default 400 m square, $N = 8$,

Table 4. Share reported within 60 s with SENTINEL components removed, and for the best baseline (named) with and without SENTINEL's fusion rule. Urban, severe: 400 m; dense: 200 m; statistics as in Table 2.

Variant	urban, N=8	severe, N=8	dense, N=50
SENTINEL	**0.57±0.05**	**0.43±0.05**	0.81±0.03
without fusion	0.52±0.05*	0.41±0.04	0.74±0.04*
without delivery (T_h=∞)	0.52±0.05*	0.42±0.05	–
without fusion, flank, delivery	0.52±0.05*	0.40±0.06	–
best baseline	0.47±0.04* (B3)	0.38±0.05 (B0)	0.79±0.03 (B7)
best baseline + fusion	0.51±0.04* (B3)	0.42±0.06 (B1)	**0.87±0.03*** (B4)

urban rubble, Poisson crashes (0.01 s^{-1}), UAVs at 30 m and 5 m/s, and a hand-set detector model (confidence vs. range and occlusion; warm-debris false alarms). Baselines B0–B8: PSO; our first version; PSO with an outage penalty; exact-λ_2 control [4]; edge potentials [5]; MAPPO [10]; sweep search; a hard 50 m nearest-neighbour wall; a responder relay chain. All were tuned with one Optuna budget on J over separate seeds (SENTINEL: $T_h = 116$ s, $\lambda_0 = 1.20$, $\beta = 1.09$); MAPPO trained to a plateau. All share seeds 0–29.

Results. SENTINEL leads in all four channel profiles, significantly against all except in the severe one (open 0.577 vs. 0.501, suburban 0.560 vs. 0.456, urban 0.575 vs. 0.475, severe 0.433 vs. 0.377). In urban rubble it reports 23–29 % more victims than the best baseline in every failure setting (Table 2). At 400 m it beats every baseline significantly from 3 to 50 UAVs (Table 3); on a denser 200 m area its mean is highest except at $N = 20$ (B7), but at no size significantly above all nine ($N = 50$: 0.809 vs. 0.792, B7).

Where the Gains Come From. Removing fusion (0.519) or delivery (0.523) lowers the urban share significantly from 0.575 (Table 4); a mesh kept connected at all times falls to 0.03–0.13. With SENTINEL's fusion rule the best urban baseline rises from 0.475 to 0.507 (B3) yet all nine stay significantly behind; in a dense 50-UAV swarm baselines with fusion reach 0.815–0.865 and B4 leads significantly.

Link Preservation. Over all runs, reliable preserved links failed within a step 69 times in 1 345 211 link-steps (5.1×10^{-5}). Without the projection (same 4 settings, seeds) violations rose from 1 in 1 663 to 42 in 1 868 link-steps, a 37-fold difference ($p < 10^{-4}$), with the on-time share unchanged. Varying λ_0 (0.3–4.8), β (0–4) or ε (0.01–0.2), or a distance spring ($R_{\max} = 30$–70 m) changes the share by ≤ 0.015, never significantly.

PX4/Gazebo and the First Scenario. 3 PX4 UAVs flew SENTINEL and B0 on 3 100 m worlds with one crash; reported victims, packet delivery and connected time agree with the fast simulator within 0.06; B0 is slightly ahead in both (0.65 vs. 0.62 in Gazebo). With the fine-tuned YOLO26n (below) in the loop on 10 such worlds, SENTINEL reports 0.89 and B0 0.91 (difference -0.02, 95 % CI -0.07 to $+0.04$, $p = 0.48$; fast simulator 0.90 vs. 0.93). In the first version's 25 m, 3-UAV scenario all methods stay connected; confirmed victims are 0.71 for SENTINEL, 0.67 for B0 (not significantly fewer) and 0.63 for the first version (significantly fewer); its retention and discovery figures are withdrawn.

Detector. On HIT-UAV [1], YOLO26n [2] matches YOLOv8n in validation mAP50 (0.948 vs. 0.947) with 19 % fewer parameters and 31 % fewer operations (TensorRT 1.81 vs. 1.73 ms on a desktop RTX 3060). On Gazebo-rendered thermal frames at 25–30 m (not real footage) it reaches 0.05 mAP50 on 5 unseen worlds, and 0.99 after fine-tuning

on 14 others, with recall 0.98 (open) and 0.96 (partly occluded) at confidence 0.40 and 0.002 false alarms per frame on warm debris. With the in-loop model fitted to these outputs (no re-tuning), SENTINEL keeps the highest mean in the six urban settings of Table 2, but leads the best baseline, B2, by only 2–7 %, never significantly.

6. Limitations and Conclusion

Results are simulated with published channel ranges and no flights; detector accuracy at 25–30 m is measured on rendered frames only, and the loop uses a detector model handset or fitted to detector outputs on rendered frames. Victims within the 5 m association gate can merge, MAPPO used one reward design, and no embedded latency was measured. SENTINEL reported more victims within 60 s than nine tuned baselines in most settings (with the fitted detector model by 2–7 %, not significantly) and was never significantly worse than any baseline in its original form. Fusion and information-triggered delivery produced the gains (in dense swarms fusion alone); the outage spring and link preservation are a safety property, cutting violations 37-fold.

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